PHY324 - Wave Propagation in Slinky

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1 Abstract

The purpose of this experiment is to investigate the features of wave propagation in a dispersive medium.
The medium being studied is a suspended slinky that is driven by a motor at speeds indicated as between 0 and 110. For this experiment, we study three chosen ranges of angular frequencies (ω), that are of interest: range#1 : $\omega>\omega_0$,range#2 : $\omega<\omega_0$, range#3 : $\omega=\omega_0$ where ω_0 is the angular frequency for a single pendulum of the slinky suspension string system.. These ranges are of interest to us since they are characteristic of the three different kinds of wave phenomena we may observe in our slinky.

Range#1 is the range of motor speeds at which we have standing waves in our slinky, range#2 is the range at which we observe exponential decay of waves (evanescent waves), and range#3 is the range at which we observe linear decay of waves.

In range #1, the relationship between the k^2 value and ω^2 is fitted to equation (4) in the introduction section where the optimized parameter values were calculated - using curve-fit in Pyhton - as $c_0 = 1.427 \pm 0.002$ m/s and $\omega_0 = 4.0 \pm 0.3$ rad/s (Ref[1]). In range #2, measured data of the amplitude of the wave along different positions established an exponential decay relationship between x (horizontal position along slinky) and y (wave amplitude) that can be estimated with (Equation 11). In range #3, measured data of the amplitude of the wave along different positions established a linear decay relationship between x (position) and y (amplitude) that can be estimated with equation (Equation 14).

2 Introduction

In this experiment, we study the features of wave propagation in a suspended slinky system driven by a motor on one end. The coil spring is 1.90±0.01m long and consists of coil loops that are attached to each other with a spring constant. The slinky system is suspended by a number of strings that are attached to each loop in the slinky (Ref [1]). Hence, each coil loop behaves as an individual pendulum.
When one coil loop is moved, it displaces the adjacent loops in either direction depending on the direction of propagation(Ref [1]). The slinky is driven by a motor that can be driven at different frequencies at one end and is fixed at the other end.

Figure 1: Our slinky wave setup, with the driving motor to the left.

The motion of the slinky wave at a point x along the slinky can be described by a sinusoidal function for the transverse wave with frequency ω along with a phase factor Φ_r. This oscillatory motion can be studied in three ranges as outlined below - ω_0 is the angular frequency for a single hanging pendulum in the slinky system.

(1)In the first range of interest $\omega > \omega_0$:

Equations (1) and (2) below fully describe the longitudinal wave motion in this range,

$$y_r = A_r \cdot cos(\omega t - kx + \Phi_r) \tag{1}$$

$$y_l = A_l \cdot cos(\omega t + kx + \Phi_l) \tag{2}$$

$$y = y_l + y_r \tag{3}$$

where $k = \frac{2\pi}{\lambda}$. And the relationship between k and ω can be described as ,

$$k^2 = \frac{\omega^2 - \omega_0^2}{c_0^2} \tag{4}$$

where ω^2 and k^2 are expected to be linearly dependent. This is assessed in this experiment since ω^2 can be measured and k^2 can be calculated using equation A5.

Also of interest in this range of angular frequency is the k and ω values at which resonant-like behavior is observed in the slinky.

Theoretically, these values are found at (Ref [1]),

$$k_n^2 = \frac{n\pi}{L} \tag{5}$$

where n = 1,2,3.. and L is the length of the suspended slinky.

And the angular frequencies at which resonant behavior occurs are,

$$\omega = \sqrt{\omega_0^2 - c_0^2 k_n^2} \tag{6}$$

This relationship is tested by visually identifying resonant behavior in the slinky and then calculating the corresponding k_n and ω values and comparing them to theoretical expectations based on equations 5 and 6

(2)In the second range of interest #2 $\omega < \omega_0$, the motion of the slinky can be described by:

$$y_r = A_r \cdot cos(\omega t + \Phi_r)e^{-kr} \tag{7}$$

$$y_l = A_l \cdot cos(\omega t + \Phi_l)e^{+kr} \tag{8}$$

$$y = y_l + y_r \tag{9}$$

In this range, if the chosen ω value is small enough, the resulting amplitudes will produce no wave motion and instead, the slinky will move in unison and amplitude will decrease as you move further from the driven end(Ref [1]).

Thus, at lower amplitudes, we can define the fixed end to be x $\equiv$ 0 and the fixed end so that y = 0 when x = 0 in the defined equation,

$$y = y_0 sin(\omega t)(e^{+kt} - e^{-kt}) \tag{10}$$

Notice that y (amplitude) takes on its maximum values when sin(ωt) = 1. Then Equation (10) becomes,

$$y = y_0(e^{+kt} - e^{-kt}) \tag{11}$$

where

$$y_0 = \frac{y_d}{(e^{+kL} - e^{-kL})} \tag{12}$$

The expression y_d characterizes the wave motion by the motor at x = L with amplitude y = $y_d sin(\omega$t).

It is expected from Equation 10 and 11 that amplitude of the wave will decrease exponentially as the wave propagates away from the driven end. This is tested below by measuring the amplitude at different positions along the slinky in order to determine an exponentially decreasing relationship.

(3)In the third range of interest $\omega = \omega_0$, the amplitude of the wave can be described by:

$$y = y_d \frac{x}{L} sin(\omega t) \tag{13}$$

Where Equation (13) is obtained by applying the boundary conditions y = 0 at x = 0 and y = $y_d sin(\omega t)$ at x = L .

Notice that y (amplitude) takes on its maximum values when $\sin(\omega t) = 1$. Then equation (13) becomes,

$$y = y_d \frac{x}{L} \tag{14}$$

From equation 14, in this range, there should be a linear relationship between the amplitude and the distance along the slinky. This is tested below by driving the slinky as an angular frequency $\omega = \omega_0$ and measuring the amplitude at different positions along the slinky.

3 Methods

The steps to follow in order to reproduce the experiment are outlined below.

(1)For the range where $\omega > \omega_0$:

Step1 - Find an estimate for c_0 by first measuring the length of the slinky. Then start a pulse off at one end of the slinky and time how long it takes for the pulse to travel to the fixed end and back. Estimate the velocity as $c_0 = \frac{2L}{time}$.

Step 2 - Find an estimate for w_0 using equation (A5), insert the measured value of the length of string L_s holding an individual coil and g = 9.8m/s

Step 3 - In this step collect data for 5 different ω and the corresponding k value.
The motor we used was a General Electric DC Motor with a Boston Gear UF109-20 AM1 Gear Reducer Box.
The motor speed can be adjusted in the range of 0 to 110 with arbitrary units.
Pick 5 motor speeds (80,85,90,95,100) so that the motor speeds are above ω_0 and spaced at even intervals.
From these motor speeds, determine the equivalent angular frequency. To do this notice that the driven motor has a rotating handle. Use a video camera to count the number of rotations of the handle per 3 seconds (at least). Multiply this value by 2π to get the equivalent angular frequency in rad/s units. This results in your ω value at that motor speed.

Now to calculate k by measuring the distance between adjacent nodes and using equation (A5) to solve for k. Plot this data on python, and use the curve-fit function from the scipy.optimize module to calculate the values of the parameters ω^2 and ω_0^2 in equation (4).

Step 4 - Calculate a conversion motor speed units to rad/s. To do so use the data collected in step 3 - ie. the motor speed and angular frequency at that motor speed - define 5 ratios: Conversion = $\frac{(motorspeed)_i}{\omega_i}$ for i = 1,2,3,4,5 the average of these 5 values can be used as the conversion factor from units rad/s to the equivalent motor speed or vice versa.

Step 5 - In this step find four ω values and the corresponding k value at four different modes of resonance. First, calculate k_n using equation (5) for n = 1,2,3,4. Then use equation (5) to calculate the corresponding ω value and use the conversion factor calculated in step 4 to convert the corresponding ω value from rad/s to motor speed units. This calculation results in the theoretical value that the motor speed should be at in order to find resonant behavior. Change the motor speed very slowly around this value, you should let the slinky be driven for at least 7 seconds to see if there is resonance and then change the speed slightly until you observe resonant behavior.

Resonant behavior is observed when the compression of the slinky is maximal (See Figure 2, obtained from Ref[2]), at which point the sound produced by the slinky is also maximal.
When resonant behavior is observed, note the motor speed and convert back to rad/s units by applying the conversion factor.

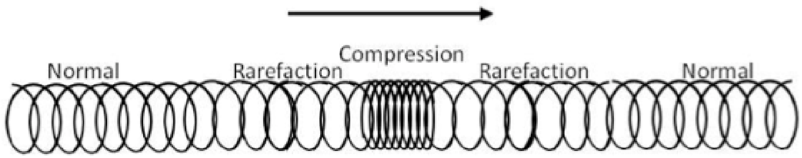


Figure 2: How we qualify resonances in our slinky.

(2)For the range where $\omega < \omega_0$:

The following steps can be followed to collect data of the amplitude of the wave at different x positions as described by equation (9).

Step 1 - using a piece of tape, mark 4 to 5 different spots on the slinky by placing the tape at the bottom of a coil loop. For this experiment the tape was placed at x = (.21m,.54m,1.21m,1.67m)$\pm$(0.01m) where the slinky length is L = 1.90 $\pm$ 0.01 m and x = L represents the oscillating end.
These lengths were chosen with sufficient spacing to observe a significant decay in the wave.

Step 2 - Drive the motor speed at 30 (equivalent to roughly 2.5$\pm$0.5 rad/s). The idea is to adjust the motor speed at a value such that the amplitude at the fixed end is approximately zero.

Step 3 - As the slinky is oscillating, visually estimate how far the tape moves to the left and right of the tape placement. To get better estimates, check how far to the left and right the tape moves and mark the point with a pen. This distance is the measured amplitude.

Step 4 - Plot the values in python and use curve-fit to fit equation (11) and find the value of y_0 and k.

(3)For the range where $\omega = \omega_0$:

The following steps can be followed to collect data of the amplitude of the wave at different x positions as described by equation (13).

Step 1 - Use the optimized value for ω_0 calculated by curve-fit in part (1) of the Methods section and multiply this value by your conversion factor found previously. Drive the motor at the resulting motor speed such that $\omega_0 = \omega$.

Step 2 - using a piece of tape, mark 4 to 5 different spots on the slinky by placing the tape at the bottom of a coil loop. For this experiment the tape was placed at x = (.21m,.54m,1.21m,1.67m)$\pm$(0.01m) where the slinky length is L = 1.9 $\pm$ 0.01 m and x = L represents the oscillating end. Now turn on the motor.

Step 3 - As the slinky is oscillating, visually estimate how far the tape moves to the left and right of the tape placement. To get better estimates, check how far to the left and right the tape moves and mark the point with a pen.

Step 4 - Plot the values in python and use curve-fit to fit equation (14) and calculate the parameter value of y_d.

4 Results

Angular Frequency (ω)	k value	Distance Between Nodes (d_n)
8 $\pm$ 1 rad/s	4.623 $\pm$ 0.005	0.6800 $\pm$ 0.0005 m
8.1 $\pm$ 0.8 rad/s	4.876 $\pm$ 0.003	0.6450 $\pm$ 0.0005 m
8.8 $\pm$ 0.9 rad/s	5.325 $\pm$ 0.004	0.5850 $\pm$ 0.0005 m
9.0 $\pm$ 0.6 rad/s	5.615 $\pm$ 0.005	0.5550 $\pm$ 0.0005 m
10.5 $\pm$ 0.7 rad/s	6.831$\pm$0.006	0.4600 $\pm$ 0.0005 m

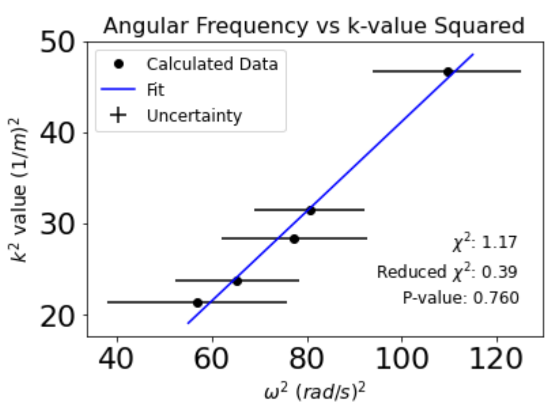
Table 1: Measured Data of Corresponding k and ω Values - see Step 3 in part (1) of the Method section.

Calculated (theoretical) k_n value	Calculated (theoretical) w value	Observed w value	Observed anti-nodes
$k_1 = 1.6534 \pm 0.0004$	4.6 $\pm$ 0.5 rad/s	4.4 $\pm$ 0.2 rad/s	1
$k_2 = 3.3069 \pm 0.0009$	6 $\pm$ 1 rad/s	6.8 $\pm$ 0.3 rad/s	2
$k_3 = 4.960 \pm 0.001$	8 $\pm$ 1 rad/s	8.6 $\pm$ 0.4 rad/s	3
$k_4 = 6.614 \pm 0.002$	10 $\pm$ 2 rad/s	10.7 $\pm$ 0.5 rad/s	4

Table 2: Measured Data of Corresponding k_n and ω Values - see step 5 in part 1 of the Method Section

	Calculated Value	Curve-fit Parameter Approximation
Single Pendulum Angular Frequency (ω_0)	1.413 $\pm$ 0.009 rad/s	1.427 $\pm$ 0.002 m/s
Constant Wave Speed (c_0)	3.457 $\pm$ .003 m/s	4.0 $\pm$ 0.3 rad/s
Amplitude Parameter (y_0)	0.0298 $\pm$ 0.0002	

Table 3: Calculated and Approximated Constant Values - ω_0 is the angular frequency for a single pendulum of the slinky suspension string system and c_0 is the wave speed (see Equation 4)
y_0 is the amplitude parameter in Equation 10, it determines the amplitudes of super-positioned waves when when resonance is observed.



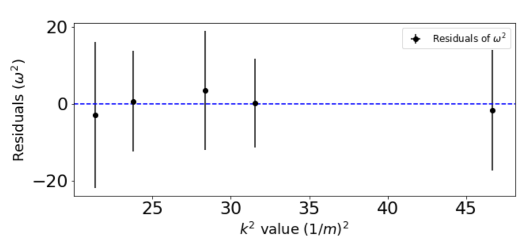


Figure 3: This graph contains the measure values of the amplitude at different positions along the slinky. Note*: the moving end of the slinky is chosen to be x = L = 1.9 $\pm$ 0.01 m and the fixed end is at x = 0 when y = 0. The fitted function is Equation (11)

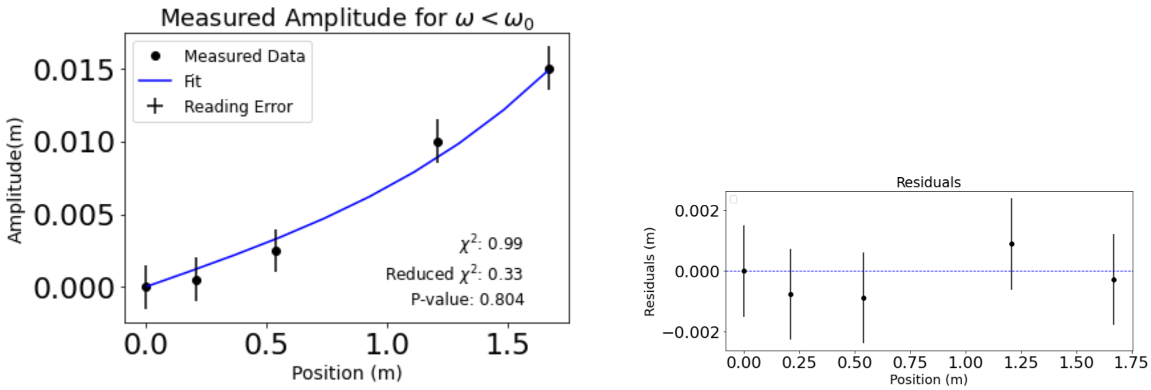


Figure 4: This graph contains the measure values of the amplitude at different positions along the slinky. Note*: the moving end of the slinky is chosen to be x = L = 1.9 $\pm$ 0.01 m and the fixed end is at x = 0 when y = 0. The fitted function is Equation (11)

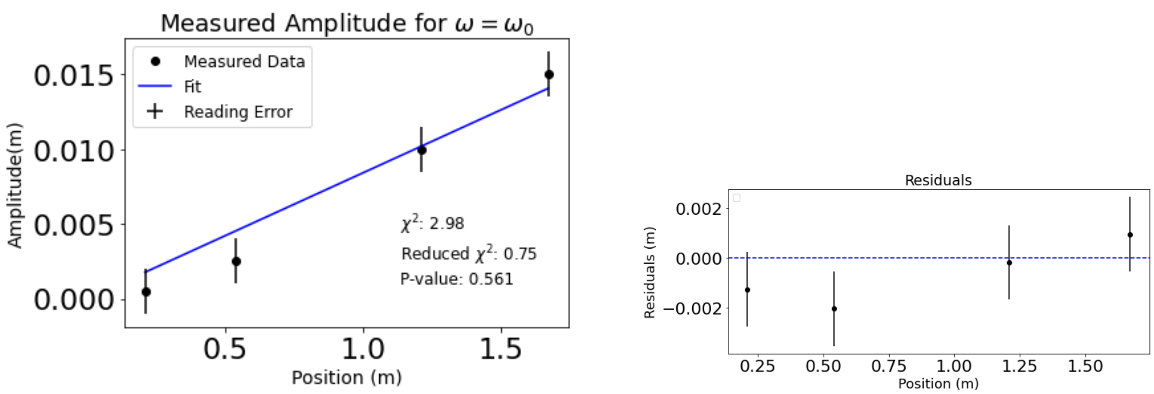


Figure 5: This graph contains the measure values of the amplitude at different positions along the slinky - see step 3 in part 2 of the methodology section. Note*: the moving end of the slinky is chosen to be x = L = 1.9 $\pm$ 0.01 m and the fixed end is at x = 0 when y = 0. The fitted function is Equation (14)

5 Discussion

Equation (4) suggests that the relationship between k^2 and ω^2 is a linearly proportional one. Qualitatively, the data points measured look to respect this relationship. Additionally, the goodness of fit assessed by the χ^2 = 1.1705 and reduced χ^2 = .3903 also suggest a reasonably good fit for the function since they are both $\approx$ 1. However, since the reduced χ^2 value $<$ 1, a better fit could be produced by collecting more data points or reducing the error in the ω^2 term in Figure 1. This is because, by the construction of the χ^2 model, χ^2 approaches 0 when the estimated uncertainties are much larger than the actual residuals. Lastly, the p-value of 0.76 also suggests that the model data is well modeled by Equation (4). Qualitatively, this is expected given that the data points lie around the fit and the residuals are a small fraction of the data values. In general, a p-value $>$ 0.05 usually suggests that the data is well predicted by a model.

The observed and theoretical values of of ω and k at angular frequencies that produce resonance are closely related as seen in Table 2. All of the observed ω values fall within the range of the theoretical values - after subtracting or adding the uncertainty value that is. This suggests that the uncertainty that result due to reading errors and structural errors in the slinky system are minimal. Lastly, the observed number of anti-nodes (positions of super positioned waved) is in agreement with the expected theoretical values.

The optimized parameters by curve-fit for ω_0 and k_0 and the calculated values are relatively close to each other. This can be seen by calculating the ratio between the calculated value and the curve-fit value for ω_0 which is $1.413/1.4273 = .98998$ and for k_0 the ratio is $3.457/4.015 = .8609$. However, the calculated values do not fall within the uncertainty range of the optimized parameter values. This is because the uncertainty in the calculated value was likely underestimated. The uncertainty was found by propagating the reading error in the timer which was $\pm$ .01s/2. This error should be at least doubled to account for the error due to the procedure followed (ie. the timer was started and stopped manually which suggests the true error is larger than just the reading error).

In the analysis of wave behavior when $\omega < \omega_0$, it is expected that as x -> 0, the amplitude will exponentially decrease as illustrated by Equation (10). As observed in Figure 2, because the driven end of the slinky is set to be x = L, the function decreases as x -> 0 which represents the fixed end of the slinky. Visually, the collected data follows this trend: the uncertainties - except for the first two data points - are small fractions of the data values and the fitted plot crosses each error bar. Additionally, the $\chi^2 = 0.99$ and the corresponding p-value of 0.804 reinforce the conclusion that the data is well predicted by the theoretical function (Equation (11)).

In the analysis of wave behavior when $\omega = \omega_0$, it is expected that as x -> 0, the amplitude will linearly decrease as illustrated by Equation (13). Similarly to the previous two ranges, the collected data follows the expected trend and the $\chi^2 = 2.98$, reduced $\chi^2 = 0.75$ and p-value = 0.561 values suggest that the data fits the theoretical function (Equation (14)).

The largest sources of error in the experiment stem from the measurements taken and their reading errors under the experiment procedure. For example, the motor speed adjustment knob does not allow for precise adjustments and therefore introduces a reading error that is propagated through calculations. Additionally, the slinky set up introduces of systematic errors as well. For example, at high angular frequencies (motor speed > 85), the slinky begins to lose its horizontally straight alignment and bends slightly.

6 Conclusion

In this report, we conducted an experiment to investigate wave propagation in a dispersive medium using a suspended slinky. We examined three different ranges of angular frequencies: $\omega>\omega_0$, $\omega<\omega_0$, and $\omega=\omega_0$.

In the first range, we measured the relationship between the wave number k and angular frequency ω. Our data indicated a linear relationship, with a good fit to the theoretical model as indicated by a p-value of 0.76. We also observed resonance behavior, which closely matched theoretical predictions - ie all the theoretical ω values all within the range of the observed values

In the second range, where $\omega<\omega_0$, we analyzed the amplitude of the wave and found that it exponentially decreased as we moved further from the driven end of the slinky. The data fitting the theoretical function supported our analysis with a p-value of 0.804.

In the third range, when $\omega=\omega_0$, we investigated wave behavior, and our measurements demonstrated a linear decrease in amplitude as we moved away from the driven end. This behavior aligned with the theoretical expectations with a p-value of 0.561.

We have therefore observed the theoretically predicted phenomena to be in close proximity with our experimental results.

Appendices

A Uncertainty Calculations

$$\Delta z = z \cdot \sqrt{\left(\frac{\Delta x}{x}\right)^2 + \left(\frac{\Delta y}{y}\right)^2} \tag{A1}$$

$$\Delta z = z \cdot \frac{\Delta x}{x} \tag{A2}$$

$$\Delta w = \sqrt{\left(\frac{w_0}{\sqrt{w_0^2 + c_0^2 k_n^2}} \cdot \Delta w_0\right)^2 + \left(\frac{c_0 k_n^2}{\sqrt{w_0^2 + c_0^2 k_n^2}} \cdot \Delta c_0\right)^2 + \left(\frac{c_0^2 k_n}{\sqrt{w_0^2 + c_0^2 k_n^2}} \cdot \Delta k_n\right)^2} \tag{A3}$$

Uncertainties in Table 2 were calculated as follows:
For calculated ω: value was calculated using equation (5) and (A3) was used to propagate errors
For observed/measured ω: the readings were observed in arbitrary motor speed units and the initial errors is the reading error for the motor speed. The values are converted (by division) to rad/s units and (A1) was applied to propagate the reading error.
For k: the reading error of L is propagated using (A1)

Uncertainties in Figure 1: the uncertainties for the k and ω values in the graph were propagated using (A2) - the equation for propagating uncertainties after squaring a term.

Uncertainties in Figure 2 and 3: the error bars represent the reading error. Errors for both axes were plotted but may be too small to see graphically.

Other equations that were used in this analysis (excluded from the introduction) are stated below:

$$\omega_0 = \sqrt{\frac{g}{L_s}} \tag{A4}$$

In range#1, the distance between two nodes (Δx) is characterized by,

$$\Delta x = \frac{\pi}{k} \tag{A5}$$

References

[1] PHY324 Wave Propagation Slinky Lab Manual (2023)
University of Toronto
- All equations are from lab manual

[2] Polarization of Waves (2023), Physics Bootcamp
`http://www.physicsbootcamp.org/polarization-of-waves.html`